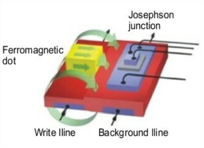

Operation of a single memory cell. The magnetic field of a superconducting write-line magnetises a ferromagnetic dot, which controls the critical current of a Josephson tunnel junction

conductors and play a major role in new technologies such as quantum information science.

Using focused helium beam to irradiate and hence disorder a nanoscale region of the superconductor to create the so-called 'quantum mechanical tunnel barrier,' researchers were able to write Josephson circuits directly into a thin film of the oxide superconductor. This direct-write method eliminated the lithographic processing and offered the promise of a straightforward pathway to quantum mechanical circuits operating at more practical temperatures.

Physicists are now collaborating with medical researchers to apply their work to the development of devices that can non-invasively measure tiny magnetic fields generated within the brain, in order to study brain disorders such as autism and epilepsy in children. In the communications field, they are developing wide-bandwidth, high-data-throughput satellite communications. In basic science, scientists are studying ceramic superconducting materials to determine the physics governing their operation, which could ultimately lead to improved materials working at even higher temperatures.

Development state

The superconducting-electronics community has seen a lot of new devices come and go without any development beyond basic characterisation. Superconductors have no electrical resistance, meaning that electrons can travel through them completely unimpeded. On the other hand, even the best standard conductors—like copper wires in phone lines or conventional computer chips—have some resistance, overcoming which requires operational voltages much higher than those that can induce current in a superconductor. Once electrons start moving through an ordinary conductor, these still collide occasionally with their atoms, releasing energy as heat. Superconductivity damps the vibrations of atoms, letting electrons zip past without collision.

Researchers are focusing on superconducting circuits made from niobium nitride, which has a relatively high operating temperature. That's become possible with liquid helium. Cheap superconducting circuits could also make it much more cost-effective to build single-photon detectors—an essential component of any information system that exploits the computational speedups promised by quantum computing.

Researchers have developed a device called as nanocryotron, after the cryotron. An experimental computing circuit developed in the 1950s, cryotron was briefly the object of interest and federal funding as the possible basis for a new generation of computers, but it was eclipsed by the integrated circuit. Superconducting circuits are used in light detectors that can register the arrival of a single light particle, or photon; that's one of the applications in which the researchers tested the nanocryotron. Researchers also wired together several circuits to produce a fundamental digital-arithmetic component called 'half-adder.'

The nanocryotron—or nTron—consists of a single layer of niobium nitride deposited on an insulator in a pattern that looks roughly like a capital 'T.' But where the base of T joins the crossbar, it tapers to only about one-tenth its width. Electrons sailing unimpeded through the base of T are suddenly crushed together, producing heat, which radiates out into the crossbar and destroys niobium nitride's superconductivity. A current applied to the base of T can thus turn off a current flowing through the crossbar. That makes the circuit a switch—the basic component of a digital computer.

After the current in the base is turned off, the current in the crossbar resumes only after the junction cools back down. Since the superconductor is cooled by liquid helium, that doesn't take long. But the circuits are unlikely to top the 1GHz speed typical of today's chips. Still, these could be useful for some lower-end applications where speed isn't as important as energy efficiency.

Their most promising application, however, could be in making calculations performed by Josephson junctions accessible to the outside world. Experiments demonstrated that currents even smaller than those found in Josephson-junction devices were adequate to switch the nTron from a conductive to a nonconductive state. And while the current in the base of T can be small, the current passing through the crossbar could be much larger—large enough to carry information to other devices on a computer motherboard.

Growing applications

With the growing demand for cheap, small, invisible, efficient, reliable, no-vibrations and non-magnetic electronics, researchers are looking towards superconducting electronics. Superconducting electronics can provide devices and circuits with properties not achievable with any other known technology; very-low-loss, zero-frequency-dispersion signal transmission lines, very-high-Q-value resonators and filters, and quantum-limited electromagnetic sensors